26900229 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCFSQ301 - DAIRY TECHNOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. The average fat content of buffalo milk is approximately: [K] / [CO1]

   a). 2.50% b). 3.50%

   c). 4.60% d). 6.60%

2. Identify the packaging method for packing sterilized milk. [K] / [CO2]

   a). Aseptic packaging b). Open packaging

   c). Returnable packaging d). Bulk packaging

3. Recognize the main purpose of concentrated sulphuric acid in the Gerber test. [U] / [CO2]

   a). To dissolve proteins and liberate fat b). To colour the fat

   c). To increase milk acidity d). To prevent bacterial growth

4. Milk obtained after altering the natural constituents of whole milk by addition, removal, exchange, and/or treatment is called: [U] / [CO2]

a). Special milk b). Raw milk

c). Skim milk d). Fermented milk

5. Select the practice that reduces the risk of cross-contamination. [U] / [CO2]

a). Using the same equipment for fermented and non-fermented products b). Appropriate separation of raw and heat-treated products

c). Mixing raw and processed products d). Storing waste with raw milk

6. The regeneration section of a plate heat exchanger primarily helps to: [U] / [CO2]

a). Increase milk acidity b). Save heating and cooling energy

c). Remove fat globules d). Sterilize the milk completely

7. Recognize how Chakka can be produced. [U] / [CO3]

a). Churning cream b). Drying milk

c). Partial removal of whey from fermented curd d). Rennet coagulation of milk

8. Choose an important quality characteristic of good makhan. [U] / [CO3]

a). Rancid flavor b). Clean flavor and smooth texture

c). Very sour taste d). Hard and dry texture

9. Idenfify the storage condition for Kheer [U] / [CO3]

a). Ambient conditions b). Refrigeration

c). Freezing d). Sun drying

10. Recognize the importance of maintaining viable probiotic cells during storage. [K] / [CO3]

a). Ensures probiotic effectiveness b). Increases milk fat

c). Prevents fermentation d). Improves milk colour

[1x10 = 10]

Part B

Shory Essay Question

Answer any **4** questions

Each question carries **5** marks

11. Comment on the freezing point and boiling point of milk. [K] / [CO1]

12. Describe the carbohydrate and mineral value of milk. [U] / [CO1]

13. List and comment on the major types of cleaning compounds used in dairy industries. [K] / [CO2]

14. Explain the steps involved in the manufacturing of cottage cheese. [U] / [CO3]

15. Explain the steps of churning and working in butter manufacturing. [U] / [CO3]

16. Discuss on Whole Milk Powder (WMP). [U] / [CO3]

[5x4 = 20]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

17. Explain the various factors affecting the transportation of raw milk in India. [U] / [CO2]

18. Describe the procedure followed for unloading milk received in cans and tankers at the dairy reception dock. [U] / [CO2]

19. Describe the manufacturing process of Yoghurt. Explain with the help of a flow diagram. [U] / [CO3]

20. Write an essay on the separation methods and processing of milk cream. [U] / [CO3]

[10x2 = 20]